Decision Tree: Transaction Fraud Classification

Planning notebook

This project is currently in the planning stage. It will teach Decision Tree classification with a deterministic synthetic transaction dataset, with particular attention to class imbalance, leakage prevention, and precision-recall trade-offs.

Educational use only: synthetic fraud labels are not evidence that a model can detect real fraud. Do not use this project for automated financial decisions.

Planned workflow

1. Generate a documented and reproducible synthetic dataset.
2. Define the target and remove post-outcome or target-derived features.
3. Create a stratified train/test split before fitting transformations.
4. Compare a controlled Decision Tree with a simple baseline.
5. Report confusion matrix, precision, recall, F1 and PR-AUC.
6. Analyse threshold trade-offs and model limitations.
7. Save and test the full preprocessing and prediction pipeline.

Status

No dataset, trained model, metrics or deployment is included yet. Those results will be added only after the implementation is complete and verified.